THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A fair test changes only one factor at a time. Reason (R): Changing several factors together makes it impossible to know which caused the effect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Animal cells burst more easily than plant cells in very dilute surroundings. Reason (R): Animal cells have a rigid cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Antibiotics are not effective against influenza. Reason (R): Influenza is caused by a virus, which has no bacterial structures for antibiotics to act on. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): An electromagnet loses its magnetism when the current is switched off. Reason (R): Soft iron does not retain magnetism. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A student studies the cooling of tea. Temperatures at 0, 2, 4, 6 minutes are 90, 78, 69, 62 degrees C. The fall during the first 2 minutes compared with the last 2 minutes is: 1

(a) greater **(b)** equal **(c)** zero **(d)** smaller

6. A bacterium of length 2 micrometres is viewed with total magnification of 1000x. Its image size is: 1

(a) 2 mm **(b)** 20 mm **(c)** 0.2 mm **(d)** 2000 mm

7. In a school of 500 students, 40 got infected with influenza in a week and 30 more next week. The number infected after two weeks is: 1

(a) 500 **(b)** 40 **(c)** 30 **(d)** 70

8. Two nichrome wires A and B are of equal length; A is thicker. The same current flows in both. Which statement is correct? 1

(a) A gets hotter than B **(b)** B gets hotter than A

(c) both stay cold **(d)** A melts first

9. A thermometer reads 3 degrees C in melting ice. A reading of 53 degrees C for warm water corresponds to a true value of: 1

(a) 56 degrees C **(b)** 50 degrees C **(c)** 53 degrees C **(d)** 47 degrees C

10. A cell loses its nucleus. The most probable result is that the cell: 1

(a) grows faster **(b)** forms chloroplasts

(c) stops functioning properly **(d)** becomes a plant cell

11. A student claims, 'I took antibiotics for my cold and it got better, so antibiotics cure colds.' The flaw is: 1

(a) colds are bacterial

(b) colds are viral and often recover by themselves

(c) antibiotics cure all diseases

(d) antibiotics are vaccines

12. A coil of wire is wound on a plastic tube and then on an iron rod, with the same current. The magnetic effect is stronger with: 1

(a) the plastic tube **(b)** equal **(c)** the iron rod **(d)** none

13. Assertion (A): An experiment should be repeated several times. Reason (R): Repetition ensures that the hypothesis is always true. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. Assertion (A): Chloroplasts are found in plant cells but not in animal cells. Reason (R): Plants make their own food while animals depend on others. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. Assertion (A): Vaccination protects a child against a disease before exposure. Reason (R): Vaccines make the body produce antibodies against the pathogen. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A student's pendulum gives 20 swings in 34 s. Find the time period and explain why 20 swings are timed. 2

17. A cell viewed at 500x magnification appears 5 mm long. Find the actual length in micrometres. 2

18. A town of 8000 has 6000 vaccinated persons. Find the percentage unvaccinated and say why it is important to keep it low. 2

19. Explain why a bird sitting on a high-voltage wire is not shocked. 2

20. Give two differences between accuracy and precision. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Critically evaluate: 'We tested only one type of soil, so all plants grow best in garden soil.' 3

22. An experiment: a student places an onion peel in concentrated salt solution. Cell contents shrink from the wall. What does this show about the cell wall and membrane? 3

23. A school finds 12 cases of jaundice in a week. List three likely causes and three preventive steps. 3

24. A student connects a coil, a cell and a switch. She places an iron nail inside the coil and the nail picks up pins. List three changes she can make to pick up more pins. 3

OR

From a graph, how do you judge whether two quantities are directly proportional? 3

25. Name the processes carried out by Amoeba that are done by organs in a human. 3

OR

Write the diseases caused by 3
(a) iodine
(b) vitamin D
(c) iron deficiency and the food source to prevent each.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Design a fair test to find which of three fabrics (cotton, wool, nylon) absorbs the most water. Describe the apparatus, variables, procedure, table and conclusion. 5

OR

Describe an experiment to study the structure of a plant cell and an animal cell, and compare the observations in a table. 5

27. In a locality, 40 people suffered from typhoid after a marriage feast. Suggest the likely source, how infection spread, and how a repeat can be prevented. 5

OR

Design an experiment to find how the number of turns affects the strength of an electromagnet. State variables, procedure, expected table and conclusion. 5

END OF QUESTION PAPER — Total: 50 Marks